

The Dirichlet Pólya Problem for Spherical Shells in General Dimension: Exact Dimension Lifting and the Shifted-Tail Bottleneck

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Abstract

For a spherical shell in dimension $d \geq 3$, the separated radial problem has Bessel orders $\ell + (d-2)/2$ and spherical-harmonic multiplicities $\kappa_{d,\ell}$. We give an exact branching decomposition of the strict shell-action proxy into positive integer- and half-integer shifted planar tails. We also prove a signed primitive comparison and retain its full positive remainder. This yields the exact identity

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} \binom{m+d-3}{d-3} D_A \left(m + \frac{d-2}{2} \right).$$

Consequently, one dimension-free positive shifted-tail inequality implies the sharp Dirichlet Pólya inequality for every spherical shell and every $d \geq 3$. The dimension lift is proved here; the inner shifted-tail inequality remains the central open lemma. We prove the outer and terminal tail regions, close the first interface-cell milestone by exact rational wall estimates, and prove all shifted tails above an explicit fixed-ratio high-energy threshold. No assertion in this manuscript treats the remaining bounded-frequency inner-tail lemma as proved.

1 Statement and status

For $0 < \rho < 1$ let

$$A_{\rho,1}^{(d)} = \{x \in \mathbb{R}^d : \rho < |x| < 1\}, \quad d \geq 3,$$

and use the strict counting convention

$$N_D(A_{\rho,1}^{(d)}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}^{(d)}) < K^2\}.$$

Write

$$\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}, \quad L_d = \frac{\omega_d}{(2\pi)^d}.$$

The desired estimate is

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq L_d |A_{\rho,1}^{(d)}| K^d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d. \quad (1)$$

For $a > 0$, define the zero-extended ball action

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z \leq a, \\ 0, & z > a, \end{cases} \quad (2)$$

put $\mu = \rho K$, and set

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_\mu(z). \quad (3)$$

For $x > 0$, let

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, [x] - 1\}.$$

In particular, $[n]_< = n - 1$ at a positive integral wall.

The only unproved statement on the primary path is the following.

Conjecture 1.1 (Positive shifted shell tails). *For $0 < \rho < 1$, $K > 0$, and*

$$r \in \tfrac{1}{2}\mathbb{N},$$

one has

$$T_A(r) := [A(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [A(r+j) + \tfrac{1}{4}]_< \leq 2 \int_r^K A(z) dz. \quad (4)$$

The zero extension makes the sum finite and makes the assertion trivial when $r \geq K$.

Sections 2–5 prove that Conjecture 1.1 implies (1) in every dimension $d \geq 3$. Section 6 records the proved tail regions, the completed first interface-cell theorem, and a fixed-ratio high-energy theorem.

2 General-dimensional spectral proxy

Put

$$a_d = \frac{d-2}{2}, \quad \nu_{\ell,d} = \ell + a_d.$$

The degree- ℓ spherical harmonics on $\mathbb{S}^{d-1}$ have multiplicity

$$\kappa_{d,\ell} = \binom{\ell+d-1}{d-1} - \binom{\ell+d-3}{d-1} = \frac{(2\ell+d-2)(\ell+d-3)!}{\ell!(d-2)!}. \quad (5)$$

Here and below a binomial coefficient with lower index larger than its nonnegative upper index is zero.

Under the unitary radial substitution

$$u(r) = r^{(d-1)/2} R(r),$$

the degree- ℓ block becomes

$$L_{\ell,d} = -\frac{d^2}{dr^2} + \frac{\nu_{\ell,d}^2 - \frac{1}{4}}{r^2} \quad \text{on } (\rho, 1), \quad u(\rho) = u(1) = 0. \quad (6)$$

Its positive eigenfrequencies are the zeros of

$$D_{\nu,\rho}(k) = J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k), \quad \nu = \nu_{\ell,d}.$$

With the standard continuous Bessel phase, let

$$\gamma_{\rho,K}(\nu) = \frac{\theta_\nu(K) - \theta_\nu(\rho K)}{\pi}.$$

The determinant factorization and phase monotonicity give the exact strict count

$$N_D(A_{\rho,1}^{(d)}, K^2) = \sum_{\ell \geq 0} \kappa_{d,\ell} [\gamma_{\rho,K}(\nu_{\ell,d})]_{<.} \quad (7)$$

The real-order phase-difference enclosure of [2, 3] gives, for $0 \leq \nu \leq K$,

$$\gamma_{\rho,K}(\nu) < A_{\rho,K}(\nu) + \frac{1}{4}. \quad (8)$$

For $\nu \geq K$, both the exact count and the action proxy vanish. Since $x < y$ implies $[x]_{<} \leq [y]_{<}$, including when y is integral, (7)–(8) imply the wall-safe proxy bound

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^<(\rho, K), \quad P_d^< := \sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell, \quad (9)$$

where

$$q_\ell = [A(\ell + a_d) + \frac{1}{4}]_{<}. \quad (10)$$

The argument above is dimension-independent after the substitutions $\nu = \ell + a_d$ and (5). In particular, the published phase enclosure is already a real-order result: no interpolation is introduced when d changes parity.

3 Exact harmonic branching

Let $p = d - 2$, so $a_d = p/2$, and define

$$r_m = m + \frac{p}{2}, \quad c_{d,m} = \binom{m + p - 1}{p - 1} = \binom{m + d - 3}{d - 3}. \quad (11)$$

Proposition 3.1 (Exact shifted-tail decomposition). *For every finitely supported sequence $(q_\ell)_{\ell \geq 0}$,*

$$\sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell = \sum_{m \geq 0} c_{d,m} \left(q_m + 2 \sum_{j \geq 1} q_{m+j} \right). \quad (12)$$

Consequently, for (10),

$$\boxed{P_d^< = \sum_{m \geq 0} c_{d,m} T_A(r_m)}. \quad (13)$$

This identity is exact at every strict action wall.

Proof. The generating functions are

$$\sum_{\ell \geq 0} \kappa_{d,\ell} s^\ell = \frac{1 + s}{(1 - s)^{d-1}}, \quad \sum_{m \geq 0} c_{d,m} s^m = \frac{1}{(1 - s)^{d-2}},$$

and

$$1 + 2 \sum_{j \geq 1} s^j = \frac{1 + s}{1 - s}.$$

Coefficient comparison gives

$$\kappa_{d,\ell} = c_{d,\ell} + 2 \sum_{m=0}^{\ell-1} c_{d,m}.$$

Substitution and finite reindexing prove (12). For (10), one has $q_{m+j} = [A(r_m + j) + 1/4]_{<}$, which proves (13). No floor has been approximated or moved. $\square$

The shifts are

$$r_m \in \begin{cases} \mathbb{N}_0 + \frac{1}{2}, & d \text{ odd}, \\ \mathbb{N}, & d \text{ even}. \end{cases}$$

Thus Conjecture 1.1 is asked only on the two grids that actually occur.

4 The signed Weyl primitive

Define the right-continuous branching staircase

$$S_d(z) = \sum_{\substack{m \geq 0 \\ r_m \leq z}} c_{d,m} \quad (14)$$

and its discrepancy primitive

$$\Delta_d(z) = \int_0^z \left(S_d(u) - \frac{u^p}{p!} \right) du. \quad (15)$$

Proposition 4.1 (Signed branching primitive). *For every $d \geq 3$ and $z \geq 0$,*

$$\boxed{\Delta_d(z) \leq 0.} \quad (16)$$

For $d = 3$, equality at positive z occurs precisely at the positive integers. For $d \geq 4$, the inequality is strict for $z > 0$.

Proof. For $0 \leq z < r_0 = p/2$, $S_d(z) = 0$, so

$$\Delta_d(z) = -\frac{z^{p+1}}{(p+1)!} \leq 0.$$

Let $M \geq 0$ and $r_M = M + p/2$. The hockey-stick identity gives

$$\int_0^{r_M} S_d(u) du = \sum_{m=0}^{M-1} (r_M - r_m) c_{d,m} = \binom{M+p}{p+1}.$$

Therefore

$$(p+1)! \Delta_d(r_M) = \prod_{j=0}^p (M+j) - \left(M + \frac{p}{2} \right)^{p+1} \leq 0, \quad (17)$$

because the arithmetic mean of $M, M+1, \dots, M+p$ is $M + p/2$.

On the half-open cell $[r_M, r_{M+1})$,

$$S_d(z) = \binom{M+p}{p}, \quad \Delta'_d(z) = \binom{M+p}{p} - \frac{z^p}{p!}.$$

If the critical point lies in this cell, it is

$$z_M^* = \left(\prod_{j=1}^p (M+j) \right)^{1/p}.$$

Direct substitution into the cell formula gives

$$(p+1)! \Delta_d(z_M^*) = p(z_M^*)^p \left(z_M^* - M - \frac{p+1}{2} \right) \leq 0, \quad (18)$$

since z_M^* is the geometric mean of $M+1, \dots, M+p$ and their arithmetic mean is $M+(p+1)/2$. If the critical point is outside the cell, the cell maximum is at an endpoint, already covered by (17). This proves (16).

For $p=1$, the cell maximum is $z=M+1$ and has value zero. For $p \geq 2$, the consecutive factors in both AM–GM applications are unequal, which gives the stated strictness. $\square$

5 Exact defect identity and conditional closure

Define

$$D_A(r) = 2 \int_r^K A(z) dz - T_A(r) \quad (19)$$

and the continuous payment

$$W_d = \frac{2}{p!} \int_0^K z^p A(z) dz. \quad (20)$$

Lemma 5.1 (Exact Weyl normalization). *For $p = d-2$,*

$$\int_0^a z^p G_a(z) dz = \frac{\Gamma((p+1)/2)}{4\sqrt{\pi}(p+2)\Gamma((p+4)/2)} a^{p+2}. \quad (21)$$

Consequently,

$$W_d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d = L_d |A_{\rho,1}^{(d)}| K^d. \quad (22)$$

Proof. Scale $z = ax$ in (2), integrate the term containing $\arccos x$ by parts, and evaluate the resulting beta integrals. This gives (21). Multiplying the difference of (21) at $a = K$ and $a = \rho K$ by $2/p!$, the duplication formula for Γ reduces the coefficient to

$$\frac{1}{2^{p+2} \Gamma((p+4)/2)^2} = \frac{1}{2^d \Gamma(d/2 + 1)^2}.$$

The last equality in (22) follows from $|A_{\rho,1}^{(d)}| = \omega_d(1 - \rho^d)$. $\square$

Proposition 5.2 (Branching-defect identity). *Let*

$$\mathcal{B}_d(A) = 2 \int_0^K (-A'(z))(-\Delta_d(z)) dz. \quad (23)$$

Then $\mathcal{B}_d(A) \geq 0$ and

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} c_{d,m} D_A(r_m). \quad (24)$$

Proof. The shell action is absolutely continuous, decreasing, and vanishes at K . Fubini gives

$$\sum_{m \geq 0} c_{d,m} 2 \int_{r_m}^K A(z) dz = 2 \int_0^K S_d(z) A(z) dz.$$

Since $\Delta'_d = S_d - z^p/p!$ almost everywhere and both endpoint terms vanish, integration by parts gives

$$2 \int_0^K S_d A = W_d + 2 \int_0^K A \Delta'_d = W_d - 2 \int_0^K A' \Delta_d = W_d - \mathcal{B}_d(A).$$

Both A' and Δ_d are nonpositive, so $\mathcal{B}_d(A) \geq 0$. Subtract (13) and use (19). $\square$

Theorem 5.3 (Conditional all-dimensional shell theorem). *If Conjecture 1.1 holds, then (1) holds for every $d \geq 3$, $0 < \rho < 1$, and $K \geq 0$. It then also holds for every shell $A_{r,R}^{(d)} = \{x \in \mathbb{R}^d : r < |x| < R\}$ and every spectral parameter.*

Proof. The case $K = 0$ is trivial, so assume $K > 0$. Every r_m in (11) lies in $\frac{1}{2}\mathbb{N}$, so the conjecture gives $D_A(r_m) \geq 0$. Equations (9), (24), and (22) yield

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^< \leq W_d = L_d |A_{\rho,1}^{(d)}| K^d.$$

For a shell with outer radius R , take $\rho = r/R$ and $K = R\sqrt{\Lambda}$ and use Dirichlet dilation. The non-strict counting form follows by applying the strict estimate at $\Lambda + \varepsilon$ and letting $\varepsilon \downarrow 0$. $\square$

6 Proved tail regions and the first interface cell

The action derivative is

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\mu}, & 0 < z < \mu, \\ \arccos \frac{z}{K}, & \mu \leq z < K. \end{cases} \quad (25)$$

Thus A is decreasing and concave on $(0, \mu)$, decreasing and convex on (μ, K) , and A, A' match at μ .

Proposition 6.1 (Outer and terminal tails). *Conjecture 1.1 holds in either of the following cases:*

1. $r \geq \mu$;
2. $G_K(r) \leq 3/4$.

Proof. If $r \geq \mu$, then $A = G_K$ on $[r, K]$; the claim is trivial when $r \geq K$. Otherwise set $B = \lceil K - r \rceil$ and zero-extend

$$g(s) = G_K(r + s) \quad (0 \leq s \leq K - r)$$

to the integer interval $[0, B]$. Since $G_K(K) = G'_K(K) = 0$, this extension is nonnegative, decreasing, convex, $1/2$ -Lipschitz, and vanishes at B . The convex shifted-floor theorem of [1] therefore bounds its ordinary-floor tail by $2 \int_0^B g$. At every sample,

$$[g(j) + 1/4]_< \leq \lfloor g(j) + 1/4 \rfloor.$$

The strict tail is consequently bounded by $2 \int_0^B g = 2 \int_r^K G_K$.

If $G_K(r) \leq 3/4$, monotonicity and $A \leq G_K$ imply $A(r + j) + 1/4 \leq 1$ for every $j \geq 0$. Every strict bracket is therefore zero. Notice that equality at 1 is harmless precisely because the bracket is strict. $\square$

We next sharpen the prescribed first milestone. For the ball action set

$$T_K(r) = [G_K(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [G_K(r + j) + \tfrac{1}{4}]_<, \quad D_K(r) = 2 \int_r^K G_K(z) dz - T_K(r). \quad (26)$$

The wall-safe translation in the preceding proof gives $D_K(r) \geq 0$.

Lemma 6.2 (A low-action $1/3$ -Lipschitz reserve). *Let g be strictly decreasing and convex on $[0, b]$, with $g(b) = 0$ and $\text{Lip}(g) \leq 1/3$. If*

$$\frac{3}{4} < g(0) < 1,$$

then

$$2 \int_0^b g(z) \, dz - \left([g(0) + \tfrac{1}{4}]_{<} + 2 \sum_{j \geq 1} [g(j) + \tfrac{1}{4}]_{<} \right) > \frac{11}{16}. \quad (27)$$

The function is understood to be zero-extended after b .

Proof. Let $h = g(0)$ and let $x \in (0, b)$ be defined by $g(x) = 3/4$. Since $g(j) < 1$, a sample contributes exactly one if and only if $j < x$. Therefore the bracketed count is

$$2[x] - 1.$$

The Lipschitz bound and $g(b) = 0$ give

$$g(z) \geq \max\{h - z/3, 0\}, \quad 2 \int_0^b g(z) \, dz \geq 3h^2 > \frac{27}{16}.$$

If $x \leq 1$, the count is one and (27) follows.

Suppose $x > 1$ and put $a = (h - 3/4)/x$. Convexity implies that, for $z \geq x$,

$$g(z) \geq \max\left\{\frac{3}{4} - a(z - x), 0\right\}.$$

Hence

$$\int_0^b g(z) \, dz \geq \frac{3}{4}x + \frac{(3/4)^2}{2a} = x \left(\frac{3}{4} + \frac{9}{32(h - 3/4)} \right) > \frac{15}{8}x.$$

Since $2[x] - 1 < 2x + 1$, the defect is larger than

$$\frac{15}{4}x - (2x + 1) = \frac{7}{4}x - 1 > \frac{3}{4} > \frac{11}{16}.$$

□

Proposition 6.3 (Exact one-interface reduction). *Assume*

$$0 < \mu - r < 1, \quad r \in \tfrac{1}{2}\mathbb{N}.$$

Define

$$n_0 = [G_K(r) + \tfrac{1}{4}]_{<} - [G_K(r) - G_\mu(r) + \tfrac{1}{4}]_{<} \in \mathbb{N}_0 \quad (28)$$

and

$$\delta_\mu(r) = \int_r^\mu G_\mu(z) \, dz.$$

Then

$$\boxed{D_A(r) = D_K(r) + n_0 - 2\delta_\mu(r)}. \quad (29)$$

Moreover,

$$2\delta_\mu(r) < \frac{4(\mu - r)^{5/2}}{15\sqrt{\mu}} < \frac{4\sqrt{2}}{15} < 1. \quad (30)$$

Consequently, the shifted-tail inequality holds if any of the following is true:

1. $n_0 \geq 1$;
2. $G_K(\max\{r, K/2\}) \geq 1$;
3. $r \geq K/2$.

The only remaining one-interface configurations satisfy

$$\boxed{r < \frac{K}{2}, \quad K < \frac{1}{\omega_0}, \quad n_0 = 0, \quad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.} \quad (31)$$

In particular,

$$r \in \left\{ \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2} \right\}.$$

Proof. Since $r + j > \mu$ for every $j \geq 1$, all samples after the first agree with the ball samples. Thus $T_A(r) = T_K(r) - n_0$. Subtracting $2 \int_r^\mu G_\mu$ from the ball integral proves (29).

The turning estimate

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}} \quad (0 < z < \mu)$$

integrates to (30), using $\mu > r \geq 1/2$ and $\mu - r < 1$. If $n_0 \geq 1$, then $D_K(r) \geq 0$ and (29) is positive.

For $r < K$, use the same integer endpoint $B = \lceil K - r \rceil$ and zero-extended function $g(s) = G_K(r + s)$. Put

$$s_* = \max\{0, K/2 - r\}.$$

Then g is $1/3$ -Lipschitz on $[s_*, B]$ and $g(s_*) = G_K(\max\{r, K/2\})$. The refined ordinary-floor theorem of [3, Theorem 3.4], followed by $[\cdot]_< \leq \lfloor \cdot \rfloor$, gives

$$D_K(r) \geq \frac{1}{2} \lfloor G_K(\max\{r, K/2\}) \rfloor. \quad (32)$$

For $r \geq K$ the same inequality is trivial. Hence the second condition and (30) prove the claim.

If $r \geq K/2$ and $G_K(r) < 1$, the translated ball action is globally $1/3$ -Lipschitz. Its tail is either zero or Lemma 6.2 applies. In the latter case $D_K(r) > 11/16 > 4\sqrt{2}/15$, so (29) is again positive. This proves the third condition.

If none of the three conditions holds, then $r < K/2$ and $G_K(K/2) = \omega_0 K < 1$, giving (31). The elementary rational bounds

$$\frac{265}{153} < \sqrt{3} < \frac{26}{15}, \quad \frac{333}{106} < \pi < \frac{355}{113}$$

(the square-root bounds follow by squaring) give $1/10 < \omega_0 < 1/9$. Consequently

$$\frac{9}{2} < \frac{1}{2\omega_0} < 5.$$

Since $r < K/2 < 1/(2\omega_0)$ and $r \in \frac{1}{2}\mathbb{N}$, only the nine displayed shifts remain. $\square$

Remark 6.4 (The residual first-interface family). Proposition 6.3 replaces the only unresolved part of the first-interface problem by a bounded nine-shift family: $2r < K < 1/\omega_0$, $r < \mu < \min\{r + 1, K\}$, and $n_0 = 0$. The inequality to be checked there is exactly

$$D_K(r) \geq 2\delta_\mu(r).$$

Proposition 6.5 and Lemma 6.6 below close this family.

Proposition 6.5 (Finite ball-wall reduction). *In the residual family of Proposition 6.3, put*

$$B = G_K(r) + \frac{1}{4}, \quad q = [B]_<, \quad e = B - q \in (0, 1],$$

and set

$$E_r(K) = D_K(r) - e.$$

Then

$$2\delta_\mu(r) < e. \tag{33}$$

Consequently, the residual one-interface inequality follows from $E_r(K) \geq 0$.

On every open K -interval containing no ball-sample wall, E_r is strictly increasing. At a wall of the first sample $G_K(r) + 1/4$, the two unit drops in $D_K(r)$ and e cancel, so E_r is continuous. At a wall of a later sample $G_K(r+j) + 1/4$, $j \geq 1$, E_r drops by two for each crossed level. It is therefore enough to check the right-hand limits at the finitely many later-sample walls, together with the first positive-count wall and any lower endpoint $K = 2r$ at which the count is already positive.

More explicitly, all walls that can occur have the form

$$G_K(r+j) = n - \frac{1}{4}, \quad j \geq 0, \quad n \in \{1, 2, 3\}, \quad r+j < 10. \tag{34}$$

Proof. Let $h = G_\mu(r)$. If $q = 0$, strict increase of $G_a(r)$ in the radius a gives $h < G_K(r) < B = e$. If $q \geq 1$, write $B = q + e$ and use the strict bracket convention; then $[B-h]_< = [B]_<$ holds exactly when $h < e$. Thus, in both cases,

$$n_0 = 0 \iff [B-h]_< = [B]_< \iff h < e.$$

When $q \geq 1$, equality $h = e$ lands on the integral wall q and causes a strict-count drop. The function G_μ is strictly convex on $[r, \mu]$, with endpoint values h and 0. It lies strictly below its endpoint chord, and therefore

$$2\delta_\mu(r) < h(\mu - r) < e,$$

which proves (33). If $T_K(r) = 0$, then $T_A(r) = 0$ and the desired inequality is immediate. Otherwise it is enough, by (29), to prove $D_K(r) \geq e$.

Away from sample walls the integer-valued tail and q are constant. Writing $x = r/K$, differentiation of the ball action and its tail integral gives

$$\pi E'_r(K) = K \left(\arccos x - x\sqrt{1-x^2} \right) - \sqrt{1-x^2}, \tag{35}$$

and

$$\pi E''_r(K) = \arccos x + x\sqrt{1-x^2} - \frac{x^2}{K\sqrt{1-x^2}}. \tag{36}$$

Here $x < 1/2$ and $K > 1$. Hence

$$\pi E''_r(K) > \frac{\pi}{3} - \frac{1}{2\sqrt{3}} > 0.$$

For $r \geq 1$, evaluation at $K = 2r$ gives

$$\pi E'_r(2r) = \frac{2\pi r}{3} - \frac{\sqrt{3}}{2}(r+1) > 0.$$

For $r = 1/2$, positivity of the tail count implies $G_K(r) > 3/4$. Since G_K is $1/2$ -Lipschitz and vanishes at K , this forces $K > 2$. Formula (35) is already positive at $(r, K) = (1/2, 2)$, since

$$\pi E'_{1/2}(2) = 2 \arccos \frac{1}{4} - \frac{3\sqrt{15}}{8} > \frac{2\pi}{3} - \frac{3}{2} > 0.$$

Equation (36) then keeps it positive. Thus E_r is strictly increasing between walls.

The stated jump rules follow directly from the weights $1, 2, 2, \dots$ in $T_K(r)$. At a first-sample wall, e equals 1 and its right-hand limit is 0, exactly matching the unit right-hand drop of D_K . Simultaneous later-sample crossings are combined at the same wall abscissa. Finally, $K < 1/\omega_0 < 10$ and $G_K(z) < K/\pi < 10/3$. Thus only $n = 1, 2, 3$ and samples $r + j < 10$ can occur, and

$$\partial_K G_K(z) = \frac{1}{\pi} \sqrt{1 - \frac{z^2}{K^2}} > 0 \quad (K > z)$$

shows that every pair (j, n) produces at most one wall. This proves the finite reduction. $\square$

Lemma 6.6 (Exact residual wall certificate). *In the residual family of Proposition 6.5, if $T_K(r) > 0$, then*

$$E_r(K) > 0. \quad (37)$$

More precisely, $E_r(K) > 3/20$ for $r \leq 4$, while for $r = 9/2$ one has

$$E_{9/2}(K) > \frac{1253}{555}.$$

Proof. Put

$$H_r(K) = 2 \int_r^K G_K(z) dz - G_K(r) - \frac{1}{4}, \quad M_r(K) = \sum_{j \geq 1} [G_K(r + j) + \frac{1}{4}]_<.$$

The first-sample bracket cancels from the definition of E_r , including at its strict walls, and hence

$$E_r(K) = H_r(K) - 2M_r(K). \quad (38)$$

For $x = r/K$, direct integration gives

$$H_r(K) = \frac{A}{2} - \frac{A \arcsin x + B \sqrt{1 - x^2}}{\pi} - \frac{1}{4}, \quad A = \frac{K^2}{2} + r^2 + r, \quad B = K \left(\frac{3r}{2} + 1 \right). \quad (39)$$

We record rational envelopes sufficient to check every wall. For $0 \leq x \leq 13/20$, let

$$a_m = \frac{\binom{2m}{m}}{4^m(2m+1)}, \quad b_m = \frac{\binom{2m}{m}}{4^m(2m-1)},$$

and set

$$L(x) = \sum_{m=0}^6 a_m x^{2m+1}, \quad U(x) = L(x) + \frac{x^{15}}{1 - x^2},$$

$$V(x) = 1 - \sum_{m=1}^6 b_m x^{2m}, \quad W(x) = V(x) - \frac{x^{14}}{1 - x^2}.$$

The power series, whose omitted positive coefficients are smaller than one, give

$$L \leq \arcsin x \leq U, \quad W \leq \sqrt{1 - x^2} \leq V.$$

Together with $333/106 < \pi < 355/113$, they imply

$$\begin{aligned} \underline{G}(K, z) &= K \left[\frac{113}{355}(W + xL) - \frac{x}{2} \right] \leq G_K(z) \\ &\leq K \left[\frac{106}{333}(V + xU) - \frac{x}{2} \right] = \overline{G}(K, z), \end{aligned} \quad (40)$$

where now $x = z/K$, and

$$\frac{A}{2} - \frac{106}{333}(AU + BV) - \frac{1}{4} \leq H_r(K) \leq \frac{A}{2} - \frac{113}{355}(AL + BW) - \frac{1}{4}. \quad (41)$$

Thus every comparison below is a rational-number comparison.

Let $k_n(z)$ denote the unique solution of $G_K(z) = n - 1/4$. Substitution in (40) gives the following pairwise disjoint brackets:

wall	bracket	wall	bracket
$k_1(3/2)$	$(22/5, 9/2)$	$k_1(2)$	$(5, 26/5)$
$k_1(5/2)$	$(57/10, 29/5)$	$k_1(3)$	$(63/10, 32/5)$
$k_1(7/2)$	$(69/10, 7)$	$k_1(4)$	$(15/2, 38/5)$
$k_2(3/2)$	$(77/10, 39/5)$	$k_1(9/2)$	$(81/10, 41/5)$
$k_2(2)$	$(83/10, 17/2)$	$k_1(5)$	$(87/10, 44/5)$
$k_2(5/2)$	$(9, 227/25)$		

(40a)

These are all later-sample walls. Indeed, the same rational bounds give

$$\frac{227}{25} < \frac{1}{\omega_0} < \frac{459}{50}$$

and, at $K = 459/50$,

$$\overline{G}(K, 11/2) < \frac{18}{25} < \frac{3}{4}, \quad \overline{G}(K, 3) < \frac{8}{5} < \frac{7}{4}, \quad \overline{G}(K, 3/2) < \frac{9}{4} < \frac{11}{4}.$$

Monotonicity in K and z excludes, respectively, every omitted level-3/4, level-7/4, and level-11/4 or higher wall.

At a right-hand wall limit, the value of M_r is the event number in the following exact order:

r	$(z, n; M_r)$
$\frac{1}{2}$	$(\frac{3}{2}, 1; 1), (\frac{5}{2}, 1; 2), (\frac{7}{2}, 1; 3), (\frac{3}{2}, 2; 4), (\frac{9}{2}, 1; 5), (\frac{5}{2}, 2; 6)$
1	$(2, 1; 1), (3, 1; 2), (4, 1; 3), (2, 2; 4), (5, 1; 5)$
$\frac{3}{2}$	$(\frac{5}{2}, 1; 1), (\frac{7}{2}, 1; 2), (\frac{9}{2}, 1; 3), (\frac{5}{2}, 2; 4)$
2	$(3, 1; 1), (4, 1; 2), (5, 1; 3)$
$\frac{5}{2}$	$(\frac{7}{2}, 1; 1), (\frac{9}{2}, 1; 2)$
3	$(4, 1; 1), (5, 1; 2)$
$\frac{7}{2}$	$(\frac{9}{2}, 1; 1)$
4	$(5, 1; 1)$
$\frac{9}{2}$	$\emptyset$.

Equations (39)–(41) then give the strict rational bounds

r	first-wall lower	first-wall upper	activation or $K = 2r$	other-wall lower
$\frac{1}{2}$	$> 3/20$	$< 8/25$	$> 11/25$	$> 13/20$
1	$> 17/100$	$< 1/2$	$> 14/25$	$> 7/10$
$\frac{3}{2}$	$> 9/25$	$< 53/100$	$> 69/100$	$> 37/50$
2	$> 2/5$	$< 29/50$	$> 37/50$	$> 39/50$
$\frac{5}{2}$	$> 23/50$	$< 63/100$	$> 22/25$	$> 81/100$
3	$> 13/25$	$< 69/100$	$> 47/50$	$> 17/20$
$\frac{7}{2}$	$> 29/50$	$< 19/25$	$> 109/100$	—
4	$> 33/50$	$< 83/100$	$> 163/100$	— .

For each $r \leq 4$, the two first-wall columns concern the right-hand value at $G_K(r+1) = 3/4$. Its displayed upper bound is smaller than both competing lower bounds, so this is the controlling wall. Formula (35) shows that H_r , and hence E_r between its later-sample walls, is increasing throughout the positive-count residual region. For $r \leq 3$ the tail is zero before its first-sample activation; for $r = 7/2, 4$ the displayed $K = 2r$ value owns the lower endpoint. Finally, for $r = 9/2$ there is no later wall and

$$E_{9/2}(9) = \frac{43}{2} - \frac{279\sqrt{3}}{8\pi} > \frac{1253}{555}.$$

This proves (37). A dependency-free replay of all 628 rational comparisons is supplied in `scripts/general_d_finite_wall_fraction_verify.py`; it is an audit of, not a numerical premise for, the displayed estimates. $\square$

Theorem 6.7 (One-interface shifted tails). *Conjecture 1.1 holds whenever*

$$0 < \mu - r < 1.$$

It also holds at the endpoint $r = \mu$ by Proposition 6.1.

Proof. The three regions in Proposition 6.3 are already proved there. In its residual family, if $T_K(r) = 0$, then $T_A(r) = 0$ and the claim is immediate. Otherwise Lemma 6.6 and Proposition 6.5 give

$$D_K(r) = E_r(K) + e > e > 2\delta_\mu(r).$$

The exact identity (29) now gives $D_A(r) > 0$. $\square$

Proposition 6.8 (Fixed-ratio high-energy tails). *For $0 < \rho < 1$, put*

$$c_\rho = \frac{\arccos \rho}{\pi}, \quad d_\rho = 1 - 2c_\rho = \frac{2 \arcsin \rho}{\pi},$$

$$\eta_\rho = G_1 \left(\max \left\{ \rho, \frac{1}{2} \right\} \right), \quad a_\rho = \frac{2\pi\rho}{1-\rho}, \quad C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

and

$$K_0(\rho) = \frac{a_\rho + 2\eta_\rho C_* + \sqrt{a_\rho^2 + 4a_\rho\eta_\rho C_* + \eta_\rho^2}}{2\eta_\rho^2}. \quad (42)$$

If $K \geq K_0(\rho)$, then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. Only $r < \mu$ requires proof. Set

$$n = \lfloor \mu - r \rfloor, \quad q = r + n, \quad B = \lceil K - r \rceil.$$

Thus $q \leq \mu < q + 1$. For integers $a < b$, write

$$\mathbf{T}(f; a, b) = \frac{1}{2} \lfloor f(a) \rfloor + \sum_{j=a+1}^{b-1} \lfloor f(j) \rfloor + \frac{1}{2} \lfloor f(b) \rfloor.$$

The strict brackets in T_A are bounded above by ordinary floors. Additivity at the split point and $A \leq G_K$ therefore give, when $n \geq 1$,

$$\frac{T_A(r)}{2} \leq \mathbf{T}\left(A(r + \cdot) + \frac{1}{4}; 0, n\right) + \mathbf{T}\left(G_K(r + \cdot) + \frac{1}{4}; n, B\right). \quad (43)$$

For $n = 0$, the first term is absent and the same inequality follows by direct pointwise majorization.

Assume first $n \geq 1$ and put

$$h(t) = A(r + t) + \frac{1}{4}, \quad m_j = \lfloor h(j) \rfloor.$$

If the nonincreasing sequence $(m_j)_{j=0}^n$ leaves its first shelf, let p be its last index on that shelf; otherwise put $p = n$. On $[0, n]$, h is decreasing, concave, and c_ρ -Lipschitz. The quantitative concave trapezoidal-floor theorem [3, Theorem 1.4], with the basic concavity bound in the no-drop case, yields

$$\mathbf{T}(h; 0, n) \leq \int_0^n A(r + t) dt + \frac{n}{4} - \frac{1 - c_\rho}{2}(n - p). \quad (44)$$

For $n = 0$, take $p = 0$ and read this as an absent identity.

Let $\sigma = -A'$. On $0 < z < \mu$,

$$\sigma'(z) = \frac{1}{\pi \sqrt{\mu^2 - z^2}} - \frac{1}{\pi \sqrt{K^2 - z^2}} \geq \kappa := \frac{1 - \rho}{\pi \rho K}.$$

Indeed, after subtracting the value at $z = 0$, the remaining bracket is decreasing in its radius parameter. Since $\sigma(0) = 0$,

$$A(x) - A(y) = \int_x^y \sigma(z) dz \geq \frac{\kappa}{2}(y^2 - x^2) \geq \frac{\kappa}{2}(y - x)^2.$$

The samples at r and $r + p$ lie on the same floor shelf, so their action drop is smaller than one. Keeping the r -term in the preceding estimate and using $r \geq 1/2$ gives

$$p < \sqrt{r^2 + a_\rho K} - r \leq \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}. \quad (45)$$

For the convex suffix, set

$$s = \max\left\{q, \frac{K}{2}\right\}, \quad t_* = s - r.$$

Then $t_* \in [n, B]$, the translated ball action is $1/2$ -Lipschitz on $[n, B]$ and $1/3$ -Lipschitz on $[t_*, B]$, and

$$G_K(s) \geq K\eta_\rho.$$

The quantitative convex theorem [3, Theorem 3.4] gives

$$\mathbf{T} \left(G_K(r + \cdot) + \frac{1}{4}; n, B \right) \leq \int_n^B G_K(r + t) dt - \frac{1}{4} \lfloor K\eta_\rho \rfloor. \quad (46)$$

The only mismatch between the two integrals on the right of (43) and the shell integral is

$$\delta = \int_q^\mu G_\mu(z) dz.$$

The turning bound from [3, Lemma 6.2] and $0 \leq \mu - q < 1$, $\mu > 1/2$, give

$$0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}. \quad (47)$$

Combining (43), (44), and (46) gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z) dz + \delta - \frac{M}{4}, \quad M = \lfloor K\eta_\rho \rfloor + d_\rho(n - p) - p. \quad (48)$$

By (45),

$$M > K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}.$$

The definition (42) is exactly the positive-root condition

$$K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} \geq C_*.$$

Thus $M > 8\sqrt{2}/15 > 4\delta$, and (48) proves the result. All inequalities use ordinary floors on the majorizing side, so strict action walls require no limiting argument. $\square$

Corollary 6.9 (A uniform small-hole sector). *If*

$$0 < \rho < \omega_0, \quad K(\omega_0 - \rho) \geq \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. For $r < \mu$, use the split (43), basic concavity on the head, and (46) with $s = K/2$. This gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z) dz + \delta - \frac{\lfloor \omega_0 K \rfloor - n}{4}.$$

Since $r \geq 1/2$,

$$\lfloor \omega_0 K \rfloor - n > K(\omega_0 - \rho) - \frac{1}{2} \geq \frac{8\sqrt{2}}{15} > 4\delta.$$

The outer tails were proved in Proposition 6.1. $\square$

7 Cell recurrence and the corrected global target

For $x_j = r + j$, set

$$Q_j = [A(x_j) + \frac{1}{4}]_<, \quad D_j = 2 \int_{x_j}^K A(z) dz - \left(Q_j + 2 \sum_{k>j} Q_k \right).$$

Then

$$D_j - D_{j+1} = 2 \int_{x_j}^{x_{j+1}} A(z) dz - Q_j - Q_{j+1}. \quad (49)$$

If

$$e_j = A(x_j) + \frac{1}{4} - Q_j \in (0, 1]$$

and

$$C_j = 2 \int_{x_j}^{x_{j+1}} A(z) dz - A(x_j) - A(x_{j+1}),$$

then

$$D_j - D_{j+1} = C_j - \frac{1}{2} + e_j + e_{j+1}. \quad (50)$$

On a fully inner cell,

$$C_j = - \int_0^1 u(1-u) A''(x_j + u) du \geq 0.$$

Individual increments in (50) can nevertheless be negative, and a negative increment need not be paid by the next floor-drop cell. Thus the viable block formulation must retain the terminal convex reserve. If J is the first index with $x_J \geq \mu$, the exact target is

$$\boxed{D_0 = \sum_{j=0}^{J-1} (D_j - D_{j+1}) + D_J \geq 0,} \quad (51)$$

where $D_J \geq 0$ is the translated convex ball-tail defect. A proof of (51) needs a quantitative, not merely nonnegative, lower bound for D_J or an equivalent matched shelf reserve.

8 A dimension-dependent no-mode region

The radial form (6), the one-dimensional Dirichlet Poincaré inequality, and $r^{-2} \geq 1$ on $(\rho, 1)$ give

$$\lambda_{\ell,n} \geq \frac{n^2 \pi^2}{(1-\rho)^2} + \nu_{\ell,d}^2 - \frac{1}{4}.$$

Since $\nu_{0,d} = (d-2)/2$, one obtains the strengthened wall

$$N_D(A_{\rho,1}^{(d)}, K^2) = 0 \quad \text{if} \quad K^2 \leq \frac{\pi^2}{(1-\rho)^2} + \frac{(d-2)^2 - 1}{4}. \quad (52)$$

The equality face belongs to the no-mode region because the counting function is strict.

9 Remaining proof obligations

The first interface-cell milestone is complete, and Proposition 6.8 proves that for each fixed ρ only a bounded frequency interval remains. The primary route now has the following precise steps.

1. For arbitrary $r < \mu - 1$, prove the shelf-block inequality (51), retaining a quantitative convex suffix reserve. The one-interface terminal block is now available with strict positive reserve, but a one-negative-cell/one-drop pairing by itself is not a valid target.
2. Insert the resulting all-frequency shifted-tail theorem into Theorem 5.3. No dimension-dependent angular or optical theorem is then needed.

If the first step fails only for an explicitly controlled family of tails, the exact identity (24) is the appropriate aggregate fallback: the weighted negative part must be compared directly with $\mathcal{B}_d(A)$.

References

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